EMERGING TECHNOLOGIES

Innovation and Disruption Agent Record

Jianbo Sun | MASY1-GC 1800 | Assignment 3 | 11 October 2026

Candidate v0.2 evaluates how generative AI could change business value. Internal retail reporting has Low scenario exposure; paid public-source briefs have High conditional exposure. These are hypothetical case findings, not observed market outcomes.

A. Agent and Version

Field	Student record
Student / team	Jianbo Sun
Assignment / specialist agent	Assignment 3 / Innovation and Disruption Agent v0.2
Date	11 October 2026
GitHub repository	https://github.com/Sun417-jianbo/special-dark
Branch	assignment-3-innovation-jianbo-20261011
Commit	aa0b4bc68a6899a7cf8a946f205bce3bbf12b2e3
Frozen scaffold version	MASY1800_ET_Agent_Scaffold_v1_0

B. Specialist Purpose and Context

Field	Student record
Governing analytical question	What innovation is proposed, and what value could it create, displace or commoditize?
Emerging technology	Transformer-based generative AI language models
Intended application	AI-drafted internal weekly sales narratives from verified tables
Organization / industry	Hypothetical midsize online clothing retailer / retail
Adoption posture	Moderate / fast-follower; contrasted with an aggressive public-source research publisher
Risk / consequence context	Incorrect narratives may misdirect inventory decisions; calculations remain in SQL/spreadsheets and humans review drafts.
Management question / time horizon	Assess innovation and business-model exposure over six months; no deployment approval.

C. Three-Level Finding

Field	Student record
ET generally	Commercially implemented offerings build on technical advances (S2-S3). Broad few-shot capability supports a qualified breakthrough interpretation relative to task-specific systems; cumulative development supports an incremental alternative. These labels do not prove business value.
ET for this application	Retail reporting is proposed incremental process innovation if it materially changes the workflow and is implemented (S1). No new foundational invention or new product for sale is established.
ET for this application in this organization	Low scenario exposure: clothing sales, customer promise and sales channel remain intact. AI changes a support activity. The public-source publisher faces High conditional exposure because clients could substitute for its core paid synthesis. The proprietary-data case warrants no final grade yet.
Bounded management implication	Investigate review-inclusive costs, factual quality and what customers pay for. Pass final adoption and implementation decisions to the relevant specialists.
Confidence / uncertainty	Moderate confidence in the exposure mechanisms; low confidence in actual savings, substitution and timing. Cases are hypothetical and have no observed deployment results.
What evidence or condition would change the conclusion	Verified implementation; customer switching and willingness-to-pay; material cost changes; evidence that proprietary data, trust or advice preserve fees.
Operational criterion	**Rule used by this specialist**
Invention and innovation	New technical capability differs from an implemented product/process. A proposal remains proposed innovation; profit is not required for the innovation label.
Entrepreneurship and commercialization	Resource organization under uncertainty differs from an actual market offering. A planned subscription does not establish demand.
Incremental and breakthrough	Improve an existing approach versus a substantial capability discontinuity against an explicit baseline. First use by a firm is insufficient.
Disruption grades	Low: support activity changes. Moderate: material activity/segment changes while value capture can persist. High: a core revenue source or advantage is directly challenged. Potentially Existential: viability threatened with weak adaptation options; require additional evidence.

D. Evidence and Testing

Field	Student record
Most important evidence/source(s)	S1: OECD implementation and novelty framework. S2: original few-shot research. S3: historical vendor API offering. S4: original experiment showing task-dependent benefits and errors. Source locations and claim limits appear under Source References and in evidence/source_register.json.
Primary test - result	Primary v0.2: proposed incremental process innovation, Low exposure. All four saved responses passed the supplied validator; all three inputs and four outputs also passed checks of the schema keywords used by the supplied contracts. All 32 original scaffold files are byte-identical.
Contrast test - what appropriately stayed stable	General technology classification remains semantically stable across cases; firm posture does not change the technology evidence. Proposed/prototype status remains distinct from implemented innovation.
Contrast test - what appropriately changed	Public-source publisher: High conditional exposure with Moderate as a credible alternative. Proprietary-data publisher: final grade withheld; Low-Moderate range, pending economics and customer evidence.
Weakness/failure found	The v0.1 publisher response passed format checks, but High lacked a clear adjacent-grade boundary and explicit early-action versus delayed-response analysis. Original instructions and response are preserved.
Revision made and why	v0.2 adds grade criteria, current/proposed business-model mapping, a stronger counterargument, both timing risks and evidence to raise or lower the judgment. The original contrast case was retained.
Remaining limitation / abstention condition	Same-session, model-authored examples; no isolated repetition, independent evaluator or business pilot. Historical task evidence cannot establish current customer substitution. The borderline case explicitly requests more information.

E. Independent Judgment and Handoff

Field	Student record
Is the agent credible enough for team consideration? Why?	My proposed assessment is that v0.2 is a useful candidate for team testing because its classifications are traceable and bounded. It is not sufficiently tested to predict market collapse or authorize deployment.
Most important recommendation to the team	Preserve the implementation gate and the chain from capability to customer substitution to value capture. Test the Moderate/High boundary on unseen cases. Pass social significance, spread, adopt/defer, readiness and live-market questions to the other experts.
AI / verification note	ChatGPT/Codex drafted the instructions, synthetic cases, outputs, report and reflection. Codex inspected the original-source sections and ran the local scripts. The student must review the evidence, independent judgment and personal reflection before submission. No independent student verification or team approval is claimed.

F. Self-Reflection

I understand the difference between technical progress and business-model change more clearly after comparing these cases. Using AI to draft an internal sales report changes a supporting activity, while using it to produce a report that customers pay for could affect the product and its price. The technology is the same, but its role in creating revenue is different.

The case involving proprietary survey data also made me question a simple assumption that AI will replace all report-writing businesses. If customers pay for information they cannot get elsewhere and for accountable advice, faster writing may support the business instead of removing its value. I need to ask what customers actually pay for before describing a technology as disruptive.

The initial response was a useful reminder that a clear label such as High still needs a clear reason. I would want to know why Moderate is less convincing and what evidence could change the judgment. AI helped produce the instructions, comparisons and checks, but those outputs do not replace my responsibility to examine the evidence. In future work, I would focus more on customer demand and review costs, and check the rubric alongside the final files so that my own reasoning and reflection are visible.

Source References

S1. OECD/Eurostat (2018). Oslo Manual 2018, Chapter 2, sections 2.2.3-2.2.5. Implementation, relative novelty and uncertain value; not a universal disruption-rating rule. https://doi.org/10.1787/9789264304604-en

S2. Brown et al. (2020). Language Models are Few-Shot Learners, abstract. Technical capabilities and limitations; no evidence of either case firm's commercial performance. https://arxiv.org/abs/2005.14165

S3. OpenAI (page updated April 24, 2024). Introducing APIs for GPT-3.5 Turbo and Whisper, opening and API section. Vendor evidence of an offering; not independent ROI or current pricing. https://openai.com/index/introducing-chatgpt-and-whisper-apis/

S4. Dell'Acqua et al. (2026). Navigating the Jagged Technological Frontier. Organization Science 37(2), 403-423. Abstract and outside-frontier results inspected. Task-specific effects do not establish customer substitution. https://doi.org/10.1287/orsc.2025.21838

Sources checked by Codex on 11 October 2026. Full verification notes and evidence boundaries are preserved in the package.